

36.3 A 0.29pJ/b 5.27Tb/s/mm UCle Advanced Package Link in 3nm FinFET with 2.5D CoWoS Packaging

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Advances in packaging technologies have accelerated heterogeneous disaggregation of SoCs by enabling the integration of IPs in different technology nodes within a single package. Such systems-in-package (SiP) require highly bandwidth-efficient, low-power die-to-die (D2D) links with low latency and BER. The Universal Chiplet Interconnect Express (UCle) standard [1] offers a blueprint on building D2D links that provide solutions to these challenges. Recently, proprietary D2D links for 2.5D interposers have been published [2-7]. However, such custom solutions limit interoperability across different vendors, which UCle standard enables. This work demonstrates a 4-to-16Gb/s/pin UCle-compliant link, implemented in a 2.5D CoWoS-S package.

Figure 36.3.1.a shows the block diagram of a single UCle advanced package (UCle-AP) PHY module, which consists of 64 TX and 64 RX data lanes, 2 clock lanes with complementary phases, a sideband (SB) clock and a SB data lane, a track lane that is used to track VT drift and a valid lane that sends a pre-determined pattern to indicate the transmission of valid data. In addition to these main signal lanes, there are 6 redundant lanes (4 data, 1 clock, 1 valid) for TX and 6 for RX, and 2 redundant lanes for SB (1 SB data and 1 SB clock) to be used to prevent yield loss if a main lane has failure. The digital portion implements a link training state machine (LTSM) that performs link initialization, calibration, and training states as outlined by the standard. Digital also includes per-lane PRBS pattern generators and checkers. Figure 36.3.1.b shows the bump map of the UCle-AP module with 10 columns. Signal exit order follows the standard requirement and minimizes lane-to-lane channel skew mismatch when connected to a partner module. To maximize BW density over the beachfront, the shoreline width spec is 388.8μm for the UCle-AP standard.

The PHY consists of single-ended half-rate NRZ transceivers and is based on a clock-forwarded, delay-matched architecture to track and minimize the impact of deterministic noise sources such as supply noise. Figure 36.3.2.a shows the transmitter top level of the UCle-AP module and consists of a common (TX_CMN) block and a TX lanes block. Each TX lane is identical to minimize mismatches. TX_CMN has a ring-oscillator-based PLL, bandgap, bias generation and two phase interpolators for data (DPI) and for clock (CPI) paths. The PLL takes a 100MHz reference clock and for 16Gb/s operation, outputs multiple phases of an 8GHz clock to both the DPI and the CPI. High-speed clock distribution is a length-matched single-ended CMOS clock tree that distributes the 8GHz CPI and DPI clocks to all TX lanes. The UCle standard requires a 16Gb/s/pin link to also support 12G, 8G and 4Gb/s/pin rates. Therefore, the PLL, clocking, TX and RX circuits support a range from 4 to 16Gb/s.

Figure 36.3.2.b shows a single TX lane. Each lane's clock path consists of a clock deskew block followed by a clock DCD sense-and-correct circuit, both of which are calibrated during link training. The clock DCD correction loop regularly updates in the background via runtime re-calibration. Each lane's data path consists of an 8:1 serializer (8:2 SER with a 2:1 MUX) followed by a programmable predriver and driver stage. Each TX lane supports at-rate pre and post driver loopback, which is critical for wafer screening for D2D links. The predriver level-shifts from core supply (VDD) to the driver supply (VCCIO). In Fig. 36.3.2.c, the driver is shown having a P+N over N CMOS architecture to enable both high swing and low swing operation, determined by the value of VCCIO. A dedicated driver supply allows the driver output stage to operate at a reduced output voltage compared to the core supply, facilitating a low-power configuration.

The RX top block consists of RX lanes and CMOS clock distribution which distributes the forwarded clock and the valid signal to the lanes as shown in Fig. 36.3.3.a. Each RX data lane consists of an unterminated analog front-end (AFE), a programmable deskew block, odd and even path samplers, a 2:8 deserializer, and supports full rate loopback mode. The clock lane features only the AFE, the output of which goes to the clock distribution. Global clock DCD is calibrated via a sense and correct mechanism during training and updates in the background via runtime re-calibration. While the TX sends complementary phases of

the clock to ensure interoperability with third party UCle receivers, in this implementation, only a single phase of the clock inside the RX is distributed to the data lanes to optimize power consumption. Programmable deskew blocks in each data lane serve as a replica to match the delay of the data path to the clock distribution path and are calibrated during training. The RX front end is a ping-pong based autozero comparator architecture as shown in Fig. 36.3.3.b. It periodically switches the incoming data between two comparator banks, thereby providing alternate precharge and evaluation cycles. By providing a regular refresh, this architecture performs periodic drift cancellation and ensures low offset error at comparator inputs. A ping pong control logic block uses a slow clock, PCLK (~100MHz), and produces control signals that enable the precharge and evaluation timing functions for the comparators, as shown in Fig. 36.3.3.c. It also minimizes the pulse-width duration of the comparator precharge cycle to improve power efficiency. Vref voltage is generated by 8b DACs that are calibrated during link training to cancel data path DCD.

Instead of a per-lane CDR, clock positioning is done via the global phase interpolators CPI and DPI in the TX and the per-lane deskew elements in TX and RX. During link training, the optimum relative positioning between CPI and DPI is recorded for each lane, and an average of these values is chosen as the global PI code. The optimum delay of each lane is then fine-tuned by calibrating the TX and RX deskew codes. VT drift during mission mode operation is tracked by utilizing the track lane as a reference and adjusting the RX deskew codes accordingly during mission mode via regular intervals of runtime re-calibration.

The UCle PHY module was implemented with 45μm micro-bump pitch and fabricated in 3nm FinFET technology. Four UCle modules were instantiated within the die to demonstrate a high bandwidth chiplet use case. Two dies were connected within a 2.5D CoWoS-S package with a ~1.4mm channel length. A D2D connection between two UCle modules consists of two links (TX+RX) and 4 modules per die has 8 links with 8.192Tb/s of data traffic. Measurements were taken for low TX swing mode (VCCIO=0.45V), with all lanes of the 8 links active with a PRBS23 pattern and 8.192Tb/s aggregate data traffic over the beachfront, to stress the links and ensure supply noise and xtalk conditions were present. During measurements, all UCle modules self-calibrated (PI, RX & TX deskew, DCC, Vref, etc. codes) via training in the digital LTSM.

RX eye scan was performed by sweeping the PI and Vref codes and was measured for 16Gb/s/pin operation. Figure 36.3.4.a shows the aggregate eye for the 64 lanes of one out of the 8 links that were active and has 0.56UI eye width (EW) and 220mV eye height (EH) opening. Vref codes were mapped to voltage based on a measured Vref DAC range of 450mV for the low-swing mode setting. Eye sweeps were measured via the per-lane PRBS generators and checkers for a PRBS23 pattern and a BER of 1e-9 due to measurement time limitations. To demonstrate the eye opening for the lower BER targets, bathtub curves were generated for the aggregate eye with measured data for 1e-10 and 1e-11 and projected data after that, as shown in Fig. 36.3.4.b. To test the impact of the noisy environment of a high data traffic chiplet, the bathtub curves are shown for two cases; when only 2 UCle links (1 module per die, 2.048Tb/s traffic) are active, and when all 8 UCle links (4 modules per die, 8.192Tb/s traffic) are active. At 16Gb/s, UCle BER spec is 1e-15. Horizontal eye opening at 1e-15 BER is 0.325UI for 2 links active and 0.29UI for 8 links active case. At BER of 1e-25 and 1e-27, the link has 0.07UI and 0.03UI opening when 8 links are active. All BER measurements are raw without FEC or CRC. Figure 36.3.5.a shows the measured aggregate eye scan for 12Gb/s/pin operation, which has an EW of 0.69UI and EH of 360mV. For 12G, 8G and 4Gb/s rates UCle BER spec is 1e-27. Measured bathtub curves in Fig. 36.3.5.b show that the link has more than 0.17UI opening at 1e-27 for 12G, 8G, and 4Gb/s/pin operations, all measured when all 8 links were active.

The link operates under a core supply (VDD) of 0.75V, high supply (VDDH) of 1.2V and TX driver supply (VCCIO) of 0.45V. One UCle module consumes a total of 297mW of measured analog power which corresponds to 0.29pJ/b power efficiency. Power efficiency breakdown across the module subsections is shown in Fig. 36.3.5.c. The link has 5.27Tb/s/mm edge bandwidth density. Latency of the link FDI to FDI, measured with a streaming controller that adds 0 latency, was 3.5ns.

Figure 36.3.6 compares the performance of this work with other state of the art D2D links and Fig. 36.3.7 shows the die micrograph where one UCle PHY module measures 388.8um×1250.08um and four modules are placed with zero-space abutment. This work achieves the best power efficiency in comparison to other demonstrated works ([3] reports power efficiency for an extrapolated configuration not demonstrated in the measured design). The BER, bandwidth density and power numbers reported in this work all meet or outperform UCle standard performance targets [1].

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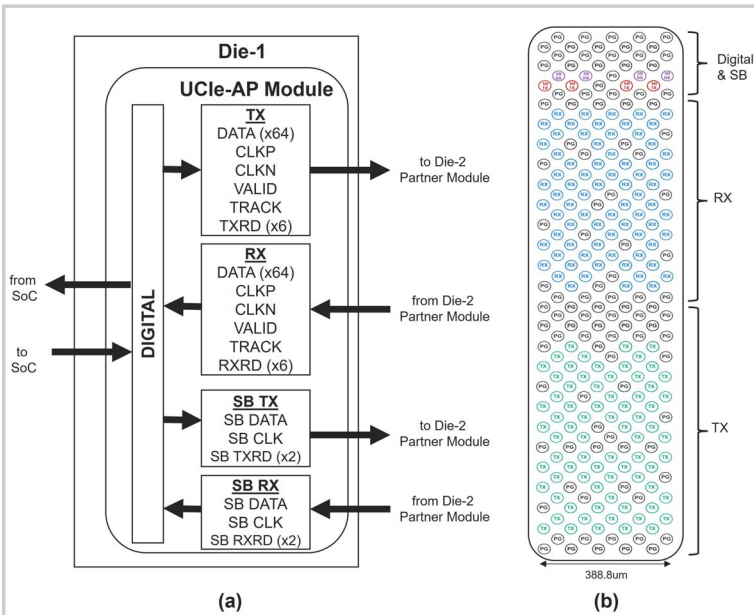


Figure 36.3.1: (a) Top level block diagram, (b) bump map of the UCle-AP module.

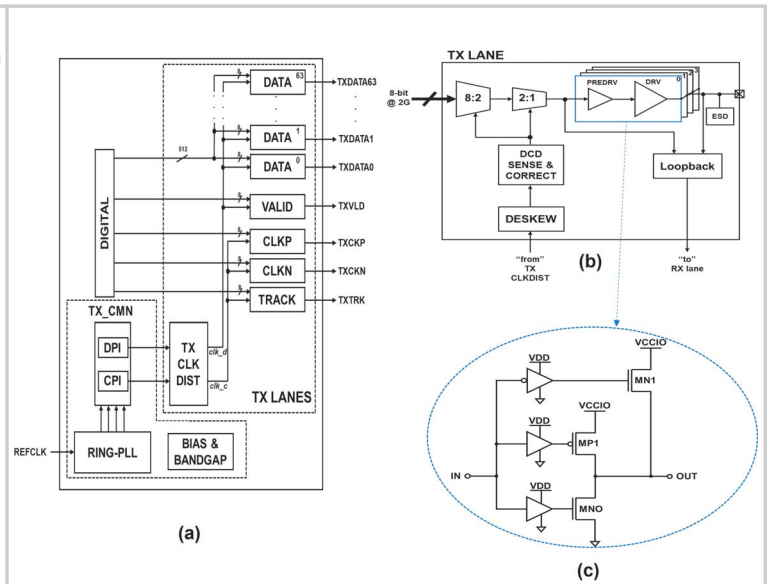


Figure 36.3.2: (a) TX top-level, (b) TX lane, (c) TX driver.

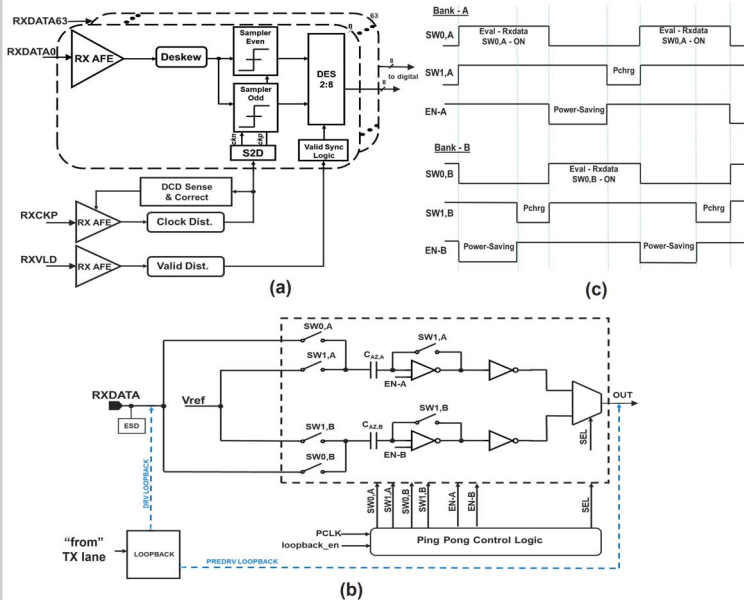


Figure 36.3.3: (a) RX top level (b) RX front-end (RX AFE) (c) timing diagram.

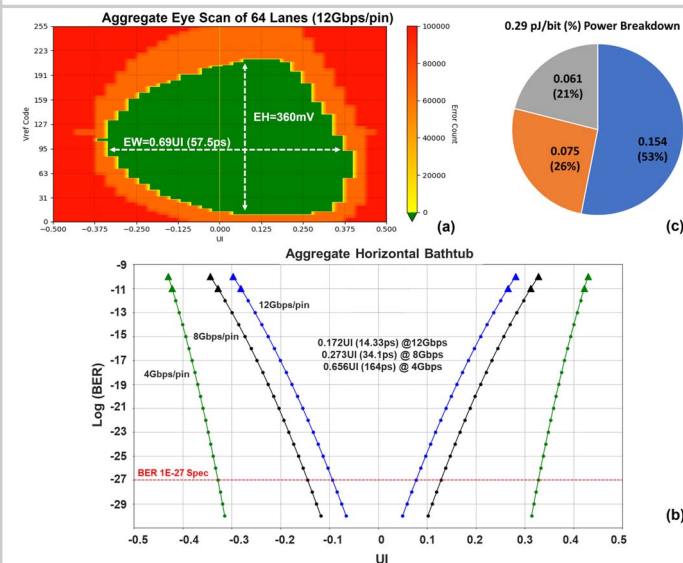


Figure 36.3.5: (a) Measured aggregate eye scan with all 8 links active at 12Gb/s/pin. (b) Measured aggregate bathtub curves at 12G, 8G and 4Gb/s/pin with all 8 links active. (c) Power efficiency breakdown.

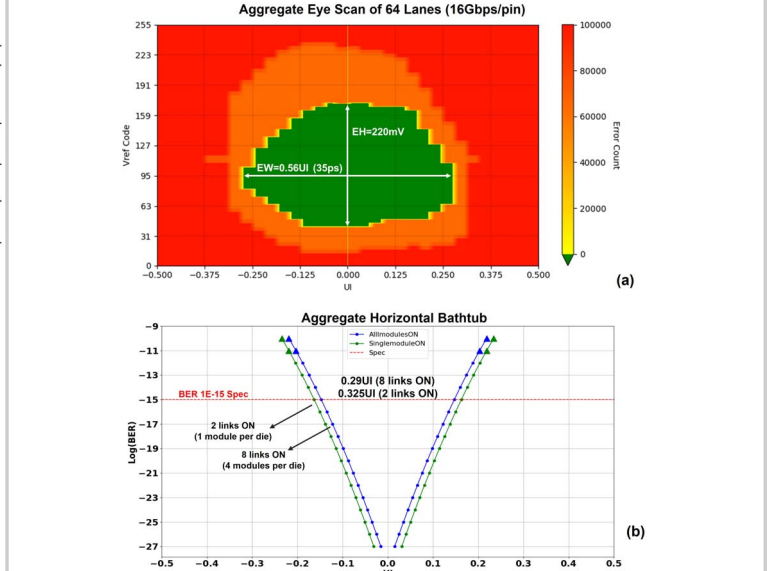


Figure 36.3.4: (a) Measured aggregate eye scan with all 8 links active at 16Gb/s/pin. (b) Measured aggregate bathtub curves at 16Gb/s/pin with 8 links and 2 links active.

	This work	VLSI 2024 [2]	VLSI 2023 [3]	ISSCC 2023 [4]	VLSI 2021 [5]	VLSI 2022 [6]	VLSI 2019 [7]
Technology Node	3nm	3nm	5nm	4nm	7nm	5nm	7nm
Standard Support	UCle	None (Proprietary)	None (Proprietary)	None (Proprietary)	None (Proprietary)	None (Proprietary)	None (Proprietary)
Channel Type	CoWoS interposer	CoWoS interposer	on-chip replica channel	Interposer	Interposer	on-chip replica channel	CoWoS Interposer
Signaling	NRZ	NRZ	NRZ	NRZ	NRZ	simultaneous bidirectional	NRZ
Channel Length [mm]	1.4	2	1.2	3	1	1.2	0.5
Bump Pitch [um]	45	-	55	50	40	55	40
Data Rate/wire [Gb/s]	16	32	25.2	32	20	50.4	8
Bandwidth of Beachfront [Tbps/mm]	5.27	3.84	5.8*	8	5.31	2.14	0.62
Power Efficiency [pJ/bit]	0.29	0.36	0.19*	0.44	0.46	0.297	0.56
FoM [Tbps/mm] / [pJ/bit]	18.17	10.67	30.5*	18.18	11.54	7.21	1.11

* Extrapolated configuration, not demonstrated in the published work

Figure 36.3.6: Performance comparison with state-of-the-art publications.

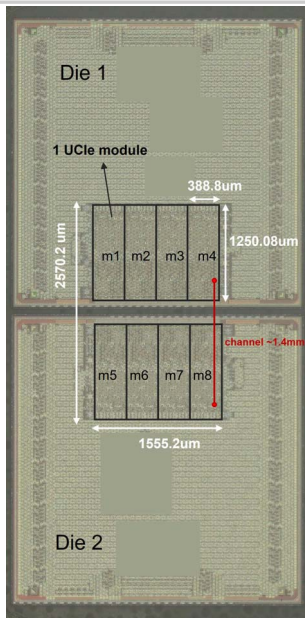


Figure 36.3.7: Package micrograph showing the die-to-die link and the UCle modules.

References:

- [1] UCle Standard Specification, <https://www.uciexpress.org/>.
- [2] J. Gu et al., "A 32Gb/s 0.36pJ/bit 3nm Chiplet IO Using 2.5D CoWoS Package with Real-Time and Per-Lane CDR and Bathtub Monitoring," *2024 IEEE Symposium on VLSI Technology and Circuits (VLSI Technology and Circuits)*, Honolulu, HI, USA, 2024, pp. 1-2.
- [3] Y. Nishi et al., "A 0.190-pJ/bit 25.2-Gb/s/wire Inverter-Based AC-Coupled Transceiver for Short-Reach Die-to-Die Interfaces in 5-nm CMOS," *2023 IEEE Symposium on VLSI Technology and Circuits (VLSI Technology and Circuits)*, Kyoto, Japan, 2023, pp. 1-2.
- [4] K. Seong et al., "A 4nm 32Gb/s 8Tb/s/mm Die-to-Die Chiplet Using NRZ Single-Ended Transceiver With Equalization Schemes And Training Techniques," *2023 IEEE International Solid-State Circuits Conference (ISSCC)*, San Francisco, CA, USA, 2023, pp. 114-116.
- [5] Y. -Y. Hsu, P. -C. Kuo, C. -L. Chuang, P. -H. Chang, H. -H. Shen and C. -F. Chiang, "A 7nm 0.46pJ/bit 20Gbps with BER 1E-25 Die-to-Die Link Using Minimum Intrinsic Auto Alignment and Noise-Immunity Encode," *2021 Symposium on VLSI Circuits*, Kyoto, Japan, 2021, pp. 1-2.
- [6] Y. Nishi et al., "A 0.297-pJ/bit 50.4-Gb/s/wire Inverter-Based Short-Reach Simultaneous Bidirectional Transceiver for Die-to-Die Interface in 5nm CMOS," *2022 IEEE Symposium on VLSI Technology and Circuits (VLSI Technology and Circuits)*, Honolulu, HI, USA, 2022, pp. 154-155.
- [7] M. -S. Lin et al., "A 7nm 4GHz Arm®-core-based CoWoS® Chiplet Design for High Performance Computing," *2019 Symposium on VLSI Circuits*, Kyoto, Japan, 2019, pp. C28-C29.